ENVS-193DS_spring-2026_final

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GitHub Repository: ENVS-193DS Spring 2026 Final

Problem 1

a.

In Part 1, my co-worker used logistic regression. We can tell because the goal is to predict a probability (will an Avocet use this microhabitat or not, which is a yes/no outcome) from a continuous measurement (distance to the water edge in meters). Logistic regression is the right tool when your response variable is binary like this.

In Part 2, my co-worker used a chi-square test of independence. This is because both variables being compared are categorical — bird species (Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). A chi-square test tells us whether the two categorical variables are associated with each other or not.

b.

Part 1: American Avocets were more likely to be found in microhabitat sites that were closer to the water's edge. This suggests that how far a spot is from the water plays a role in whether or not an Avocet chooses to use it.

Part 2: The three bird species did not use the five habitat types equally. Each species tended to show up more in certain habitats than others, meaning the habitat a bird is found in is related to which species it is.

c.

One additional variable that could be worth looking at is water depth, measured in centimeters at each survey point using a ruler or depth probe. American Avocets are wading birds that forage by sweeping their bill through shallow water, so sites that are too deep or too dry may simply not be usable for feeding, which would directly affect whether or not Avocets choose to use a given microhabitat.

Problem 2

a.

```r
nests_clean <- nests |>
  select(case_control,
         height_cm,
         ant_species) |>
  filter(ant_species == "AB" |
         ant_species == "RRB" |
         ant_species == "BBR") |>
  mutate(ant_species = factor(ant_species,
                              levels = c("AB",
                                         "RRB",
                                         "BBR"))) |>
  mutate(full_name = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  ))

str(nests_clean)
```

```
'data.frame':	270 obs. of  4 variables:
 $ case_control: int  1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm   : num  392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ full_name   : chr  "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae"
```

```r
slice_sample(nests_clean, n = 10)
```

```
   case_control height_cm ant_species              full_name
1             0     167.0         RRB   Crematogaster mimosae
2             0      76.5         RRB   Crematogaster mimosae
3             0     264.5         RRB   Crematogaster mimosae
4             0     236.0         RRB   Crematogaster mimosae
5             1     234.5         BBR Crematogaster nigriceps
6             0     163.5         RRB   Crematogaster mimosae
7             0     279.0         RRB   Crematogaster mimosae
8             0     185.0         RRB   Crematogaster mimosae
9             0     207.0         RRB   Crematogaster mimosae
10            0     155.5         RRB   Crematogaster mimosae
```

b.

The 1s and 0s in the case_control column represent whether a given tree contained a bird nest (1) or was a nearby available tree with no nest present (0). This is a binary outcome used to compare the characteristics of trees that birds actually chose to nest in versus trees that were available but unused.

c.

Tree height (height_cm, continuous variable measured in centimeters): As tree height increases, the probability of nest occurrence would likely increase because taller trees may offer more structural complexity and concealment, making them more attractive and safer nesting locations. The authors note that all nests found in the study were located in live trees more than 1.5 m in height, suggesting a minimum height threshold for nest use. (Methods, paragraph 2; Results, paragraph 1)

Acacia ant species (ant_species, categorical variable with 3 levels: C. sjostedti, C. mimosae, and C. nigriceps): The ant species occupying a tree could influence the probability of nest occurrence because different species vary in how aggressively they defend their host trees against predators. C. mimosae and C. nigriceps are aggressive defenders against both mammalian and insect herbivores, and birds nesting in those trees may benefit from protection against nest predators. In contrast, C. sjostedti provides only minimal protection, making those trees potentially less safe or attractive for nesting birds. Additionally, C. nigriceps prunes the apical buds of its host tree, creating a denser canopy that likely provides extra concealment from predators such as snakes, mesocarnivores, and raptors. (Introduction, paragraphs 2 and 4; Discussion, paragraphs 2 and 3)

d.

Model Number
Model 1
Model 2